UNIT 1 · KINEMATICS · MAKE-UP QUIZ

Make-Up Quiz — Show the Process

Physics 2026–27 · 15 minutes · 10 points · replaces your Mini Quiz score if it is higher

NAME _____ DATE _____

Two problems, one equation each. The process is what earns the points: **K** 1 point, **D** 1 point, **E** 1 point, **A** 2 points. For each problem, fill in the three values you are given, write **?** for the one you are asked to find, and **cross out** the one the problem never mentions. Then use the equation that does not contain the crossed-out variable. No notes; a calculator is fine.

$v_f = v_i + at$ · $\Delta x = v_i t + \frac{1}{2}at^2$ · $v_f^2 = v_i^2 + 2a\Delta x$ · $\Delta x = \frac{1}{2}(v_i + v_f)t$

1 The cart on the ramp

KDEA · 5 POINTS

A lab cart is released from rest and rolls down a ramp with a constant acceleration of $0.40\ m/s^2$. How long does it take to travel $0.80\ m$?

v_i = _____ v_f = _____ a = _____ Δx = _____ t = _____

K	DONE ABOVE: THREE VALUES FILLED IN, WITH UNITS
D	THE TARGET & ITS UNIT
E	THE EQUATION WITHOUT THE CROSSED-OUT VARIABLE
A	SYMBOLS FIRST · THEN NUMBERS WITH UNITS · BOX THE ANSWER

2 Braking to a stop

KDEA · 5 POINTS

A car moving at $10\ m/s$ brakes with constant acceleration and comes to a stop in $25\ m$. Find its acceleration, sign included.

$v_i =$ _____ $v_f =$ _____ $a =$ _____ $\Delta x =$ _____ $t =$ _____

K	DONE ABOVE: THREE VALUES FILLED IN, WITH UNITS
D	THE TARGET & ITS UNIT
E	THE EQUATION WITHOUT THE CROSSED-OUT VARIABLE
A	SYMBOLS FIRST · THEN NUMBERS WITH UNITS · BOX THE ANSWER